Fred Hutch Cancer Center

Translational Data Deep Dive

Building Computational Workflows

March 11th, 2026, DaSL Lounge and Teams

What is a Deep Dive?

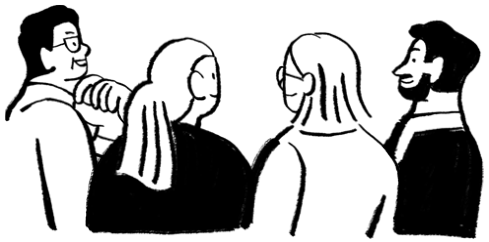
A hands-on, interactive quarterly series to help build practical data science skills at Fred Hutch

Learn Concepts

*Understand the **what**, **why**, and **when** behind data science approaches.*

See the Tools in Action

***Live demonstrations** using Hutch-supported tools and infrastructure.*



Ask & Engage

These sessions are interactive ask questions in person or online using Slido.

Get your hands dirty!

Try things yourself during the co-working time

Who we are?

And why should we be teaching you about computational workflows?

Sita: I biologist by training, I have used WDL's for 5 years for analysis of bulk genomic and single-cell data.

Taylor Firman: A bioinformatics software engineer that has used Bash/SLURM/Python in biophysics & genomics, but is much happier having discovered WDL and Docker.

Emma Bishop: A bioinformatician who learned workflows to run single-cell pipelines in the cloud and now uses WDL instead of bash scripts to analyze T cells.

Who else is here to help today

Scientific Computing

- Dan Tenenbaum, Scientific Computing
- Michael Gutteridge, Scientific Computing

PROOF Workbench

- Scott Chamberlain, OCDO
- Sean Kross, OCDO

Training Team

- Chris Lo, OCDO
- Ted Laderas, OCDO

Cirro

- Sam Minot, Cirro

Before we begin…

Some housekeeping to make this session work for everyone

- You can find the materials associated with this Deep Dive @
  - Deep Dive Pages
  - GitHub

- We will be using Slido
  - Just go to slido.com and enter this code #8758973
  - More details on how to use Slido here

- We follow the Training Teams DaSL Participation guidelines

Learning Objectives

Recognize When Workflows Are Useful

*Learn **when** a computational workflow is the right solution for your analysis*

Don't Build Workflows from Scratch

*Know about how you can leverage resources like the **WILDS WDL Library** to assemble your own workflow*

Understand the Structure of a WDL

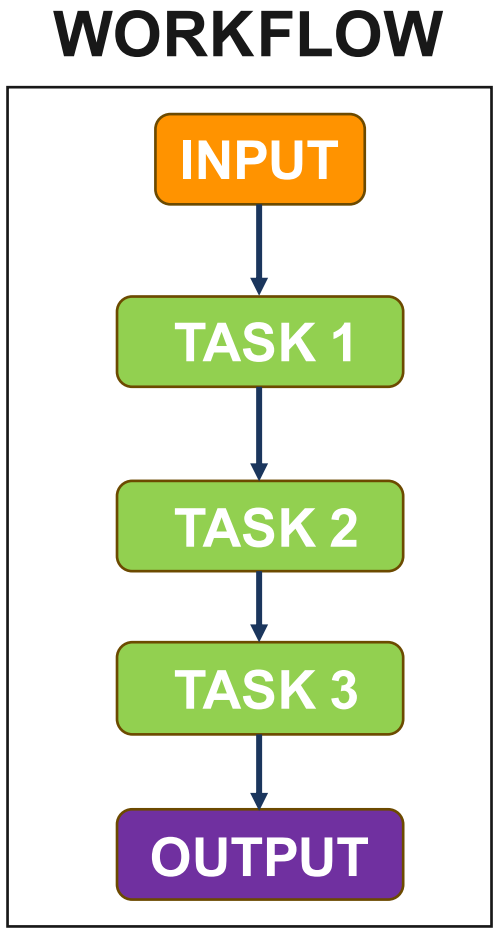
*Learn how to read and interpret the key components of **WDL**, a language to write workflows in*

Learn how to run workflows at Fred Hutch

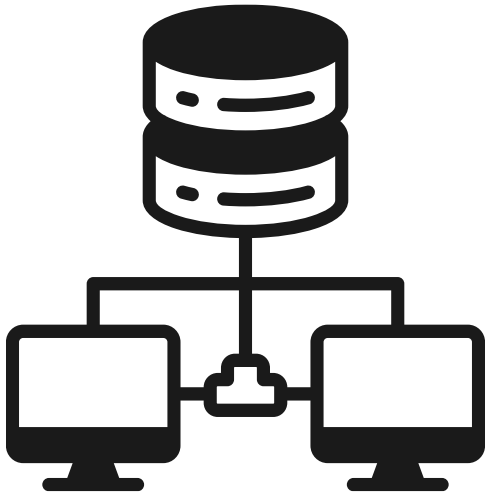
*See how workflows can be run using **PROOF Workbench** and Hutch infrastructure*

What is a Workflow?

A workflow describes all the steps needed to turn input data into results.



What are the advantages of writing and running a workflow?

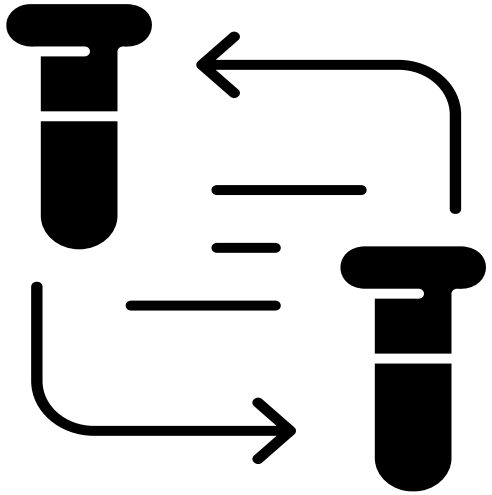


Scalable

Many samples

↓

Parallel processing

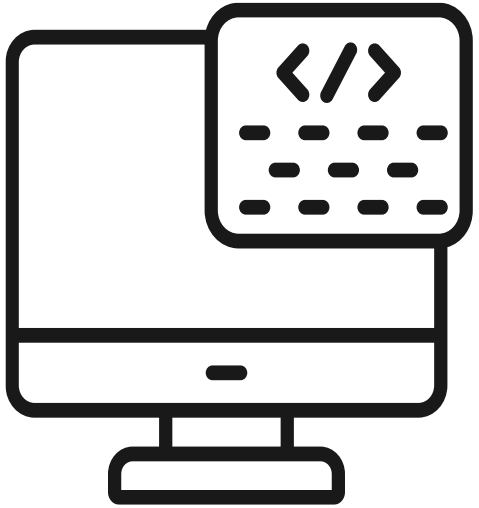


Reproducible

Defined steps

↓

Same results



Shareable

Reusable across teams

When to use a workflow?

Make a Workflow When…	DON’T Make One When…
It’s **repetitive**	You’re **still figuring things out**
It’s **multi-step**	It’s **exploratory**
It’s **multi-sample**	It’s **tiny**
It needs to **scale**	It’s **single-step**
It needs to be **reproducible**	It’s **temporary**

So how do you write a workflow?

There are special workflow languages (code) in which you can give instructions… we like

Readable

workflows look like step-by-step instructions

Modular

tools are separated into reusable tasks

Built for genomics pipelines

designed for large bioinformatics analyses

How do I get started with WDL workflows?

You will usually need 3 files

Ingredients

What data to use

`inputs.json`

Specifies **inputs**

File paths (FASTQs, BAMs, etc.)

Defines any input values for a task

Equipment

What resources to use

`options.json`

Specifies memory to use

What compute to use (HPC, Cloud, etc.)

Specifies where outputs go

The Recipe

What tasks to run

`workflow.wdl`

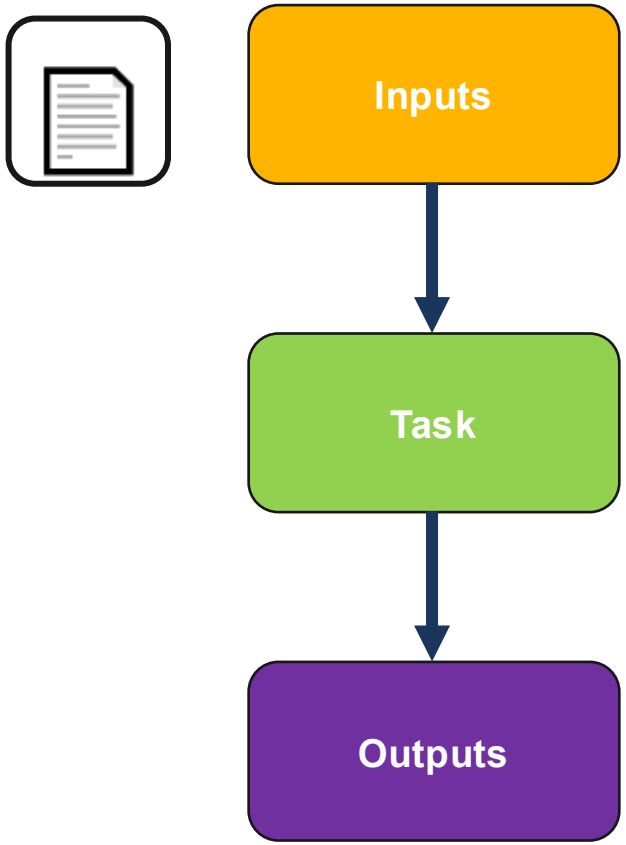
Defines the tasks

Connects inputs → tasks → outputs

Declares dependencies between steps

hello_world **Workflow**

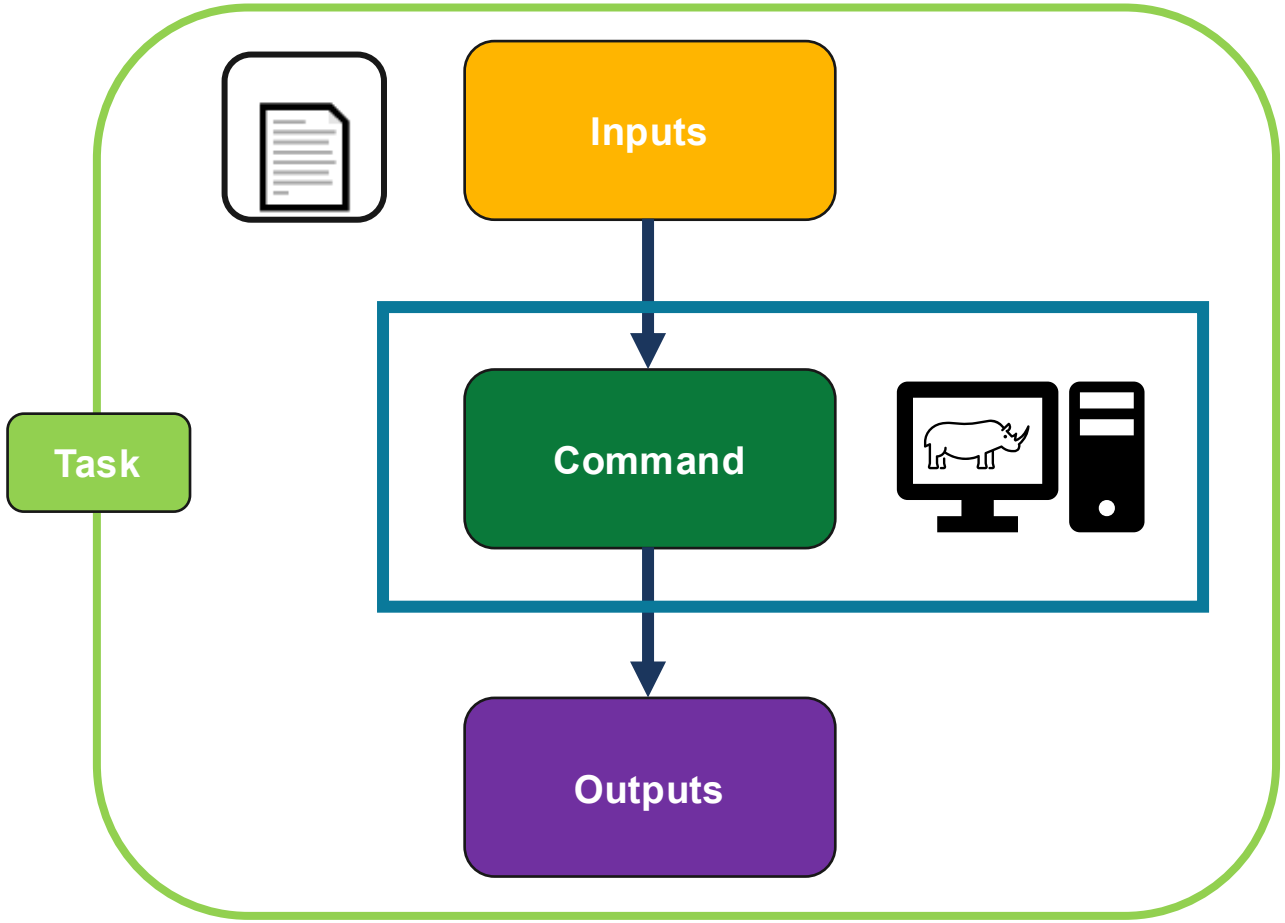
Human-readable overview of what's happening




```
workflow hello_world {
  meta {
    description: "Writes a personalized greeting"
    outputs: {
      out_file: "A file containing a greeting"
    }
  }

  parameter_meta {
    your_name: "Name to greet"
    your_number: "Your favorite number"
  }

  input {
    String your_name = "Fred"
    Int your_number = 29
  }

  call greeting {
    input:
      name = your_name,
      number = your_number
  }

  output {
    File out_file = greeting.out
  }
}
```

“Hello World” Task: greeting

Very similar vibes, but at the “building-block” task level:

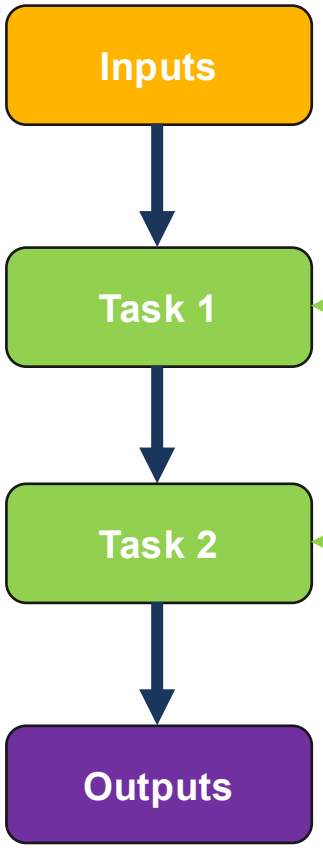



```
task greeting {
  meta {…
  }

  parameter_meta {…
  }

  input {
    String name
    Int number
  }

  command <<<
    echo "Hello, ~{name}!" > "~{name}.txt"
    echo "Your favorite number is ~{number}" \
      >> "~{name}.txt"
  >>>

  output {
    File out = "~{name}.txt"
  }

  runtime {
    docker: "getwilds/awscli:2.27.49"
    memory: "2 GB"
    cpu: 1
  }
}
```

A real-world example: ww-sra-salmon

Benefits of WDL become clear at scale…

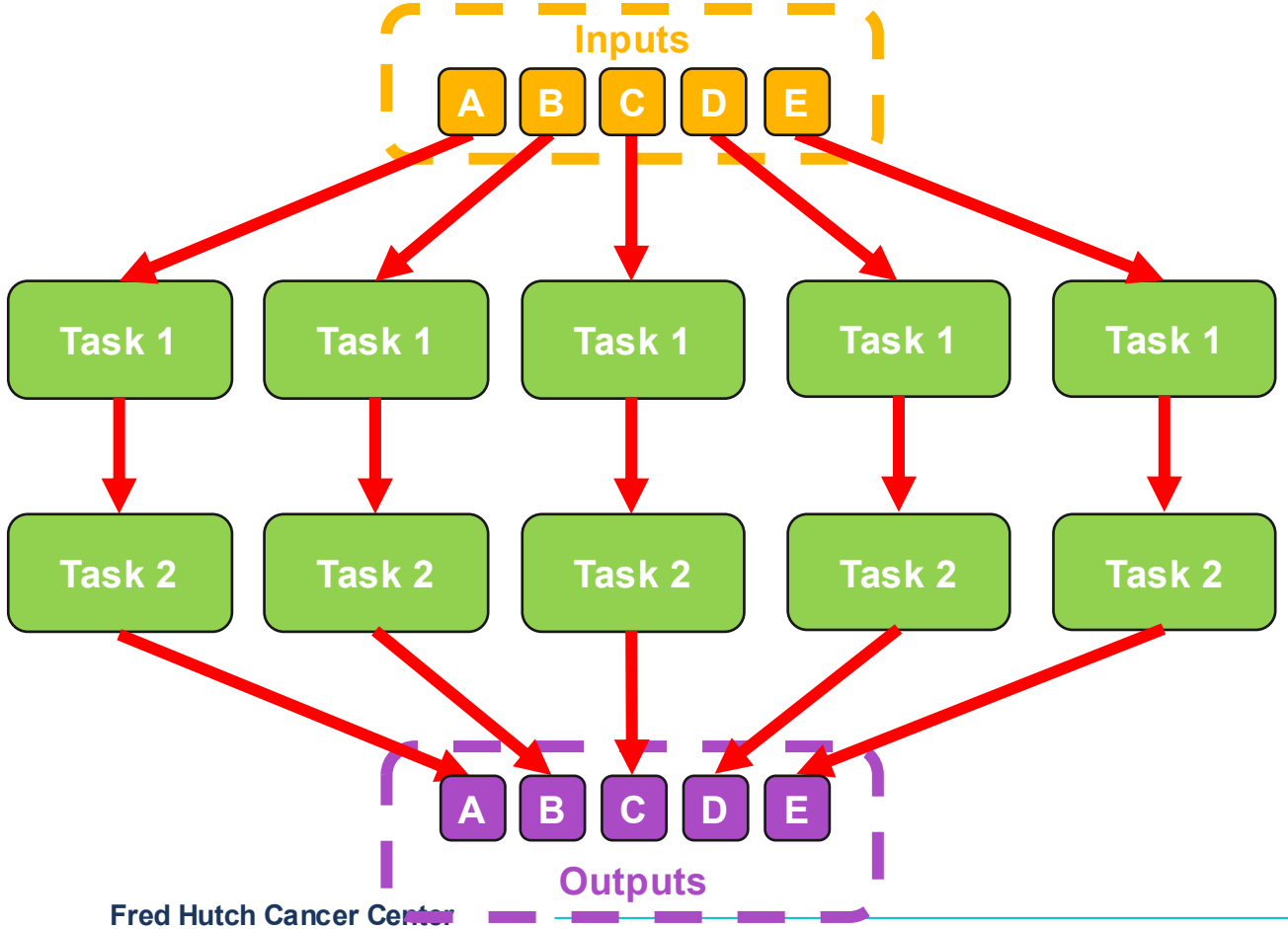
1. Specify what data I'm interested in
2. Download data from source
3. Post-process as necessary
4. Save final results locally




```
import modules/ww-sra/ww-sra.wdl as sra_tasks
import modules/ww-salmon/ww-salmon.wdl as salmon_tasks

workflow sra_salmon {
  input {
    Array[String] sra_id_list
    File salmon_index_zip
  }

  # Download FASTQ files from SRA
  call sra_tasks.fastqdump { input: sra_id = id }

  # Quantify each sample with Salmon
  call salmon_tasks.quantify { input:
      salmon_index_dir = salmon_index_zip,
      fastq_r1 = fastqdump.r1_end,
      fastq_r2 = fastqdump.r2_end,
  }

  output {
    Array[File] quant_dirs = quantify.salmon_quant_dir
  }
}
```

A real-world example: ww-sra-salmon

Benefits of WDL become clear at scale…




```
import modules/ww-sra/ww-sra.wdl as sra_tasks
import modules/ww-salmon/ww-salmon.wdl as salmon_tasks

workflow sra_salmon {
  input {
    Array[String] sra_id_list
    File salmon_index_zip
  }

  # Scatter across nodes for max parallelization
  scatter ( id in sra_id_list ){

    # Download FASTQ files from SRA
    call sra_tasks.fastqdump { input: sra_id = id }

    # Quantify each sample with Salmon
    call salmon_tasks.quantify { input:
        salmon_index_dir = salmon_index_zip,
        fastq_r1 = fastqdump.r1_end,
        fastq_r2 = fastqdump.r2_end,
    }
  }

  output {
    Array[File] quant_dirs = quantify.salmon_quant_dir
  }
}
```

Customize the workflows as needed

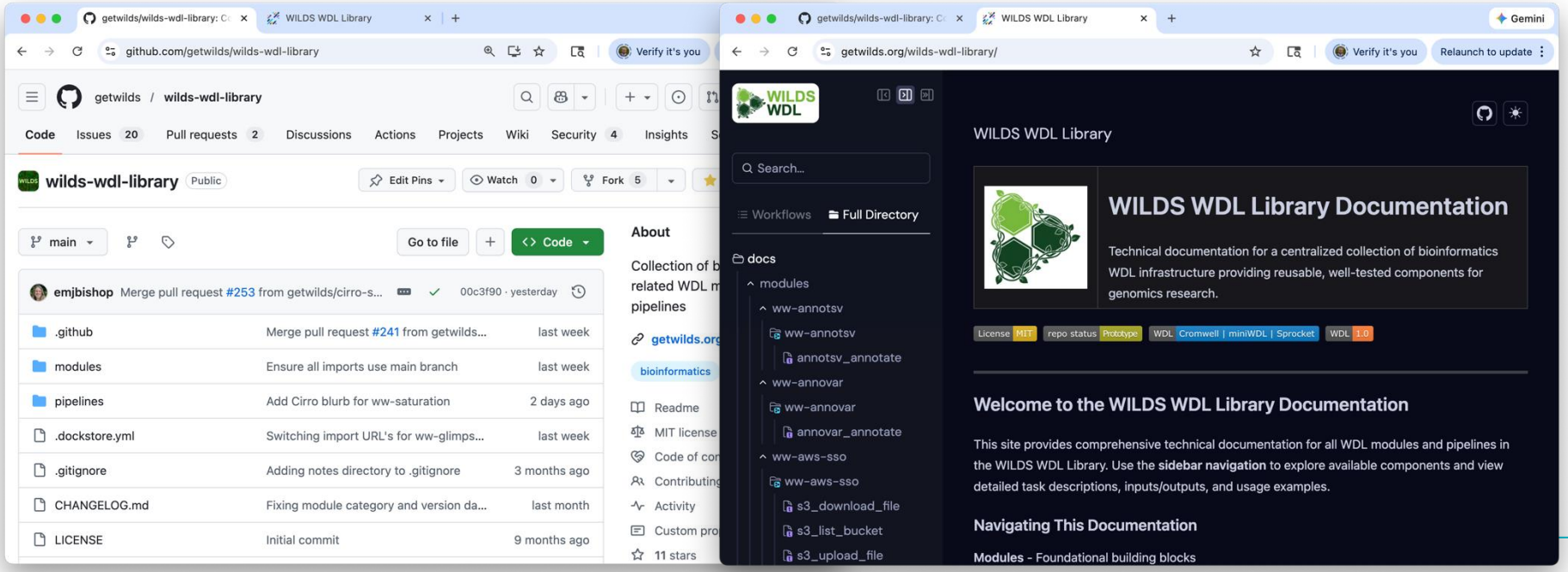
Straightforward to customize a workflow to suit your research needs

Problem	Solution	Bioinformatics Example
I want to provide my own datasets as input…	Update the inputs to accept a list of files!	Array[File] data type [file1.txt, file2.txt, …]
I'd prefer to use a different tool for post-processing…	Swap out the import for whichever one you prefer!	ww-salmon → ww-star DNAseq → ww-bwa + ww-gatk
I need to add an extra post-processing step…	Import another module and tack a task onto the bottom!	RNAseq → ww-deseq2 QC → ww-fastqc
I want to save my results to an S3 bucket…	You guessed it… there's a module for that!!	ww-aws-sso (Upload & Download)
I don't see a module that fits my needs…	We'll help you build one!!!	WILDS WDL Library WILDS Docker Library

Easiest way to write a WDL: use someone else's!

Plenty of existing WDL workflows out there… including right here at Fred Hutch!

- WILDS WDL Library: curated collection of pre-built, well-tested, well-documented workflows
  - We can help you build one via the WILDS WDL Development Program!!



More details here!!!

Other Resources

Broad's WARP
GATK Workflows
BioWDL
Dockstore

Let’s see workflows in action!

Demo: hello-world.wdl

FredHutch / data-deep-dives

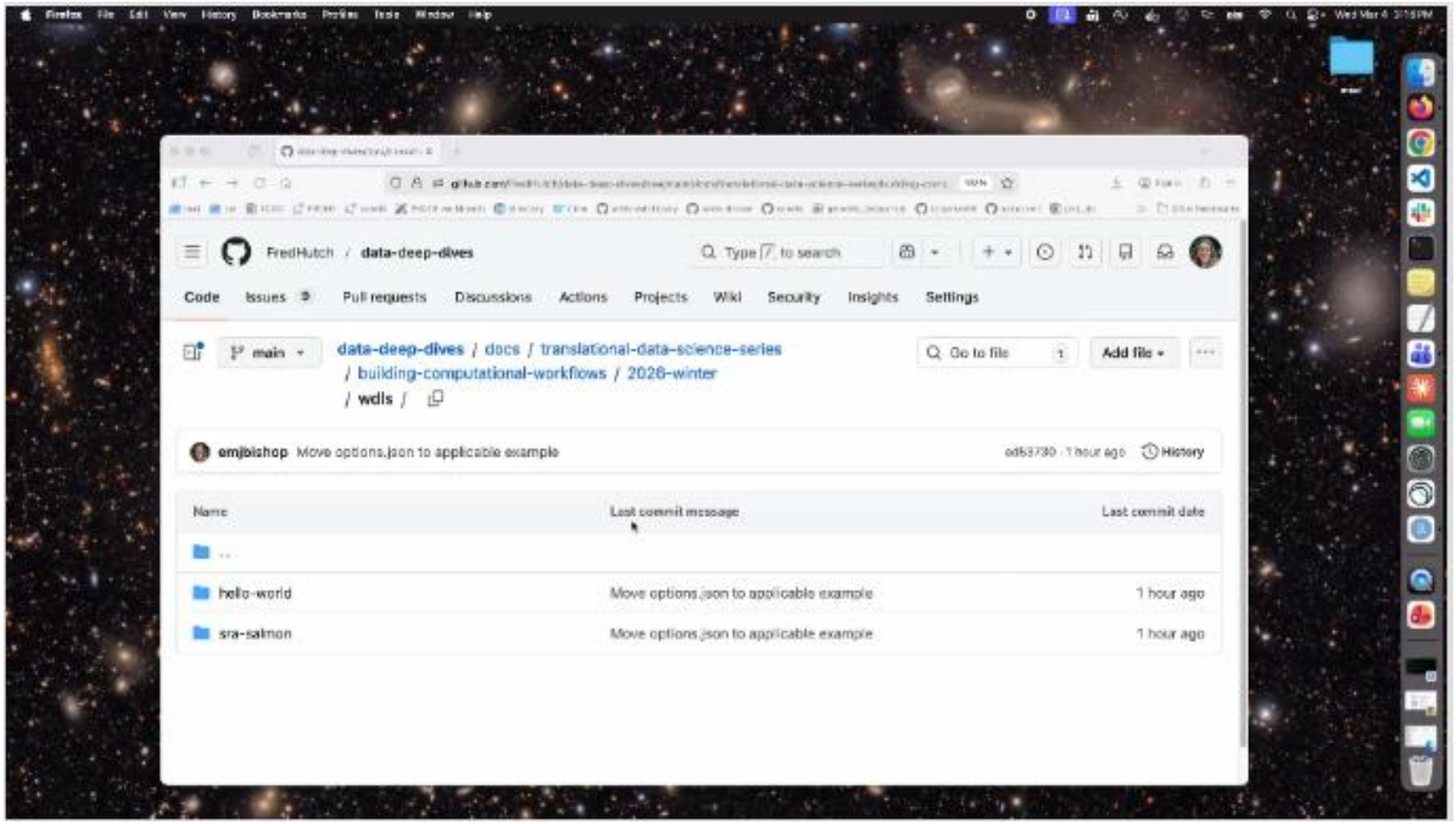
Code | Issues 9 | Pull requests | Discussions | Actions | Projects | Wiki | Security | Insights | Settings

main — data-deep-dives / docs / translational-data-science-series / building-computational-workflows / 2026-winter / wdls / hello-world /

sitapriyamoorthi Rename docs/translational-data-science-series/building-computational-... 2339748 · 4 hours ago History

Name	Last commit message	Last commit date
..		
hello-world-inputs.json	Rename docs/translational-data-science-series/building-computat...	4 hours ago
hello-world.wdl	Rename docs/translational-data-science-series/building-computat...	4 hours ago
options.json	Rename docs/translational-data-science-series/building-computat...	4 hours ago

Demo: sra-salmon.wdl

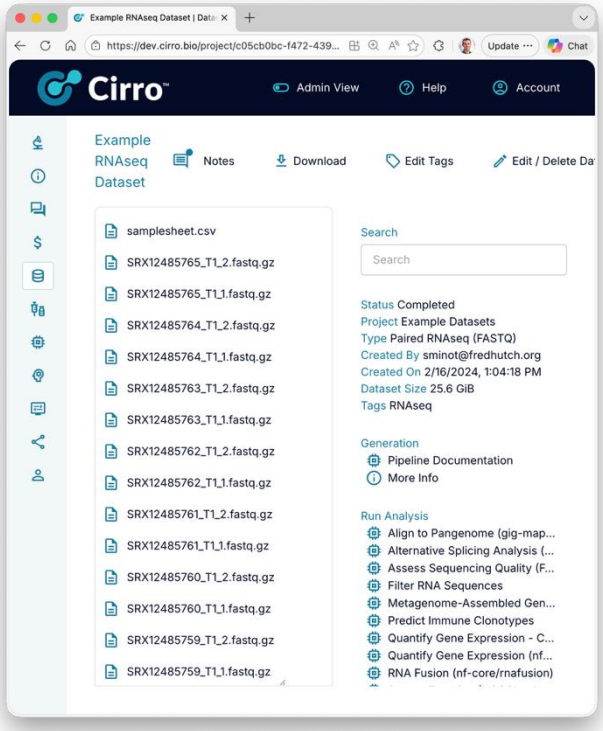


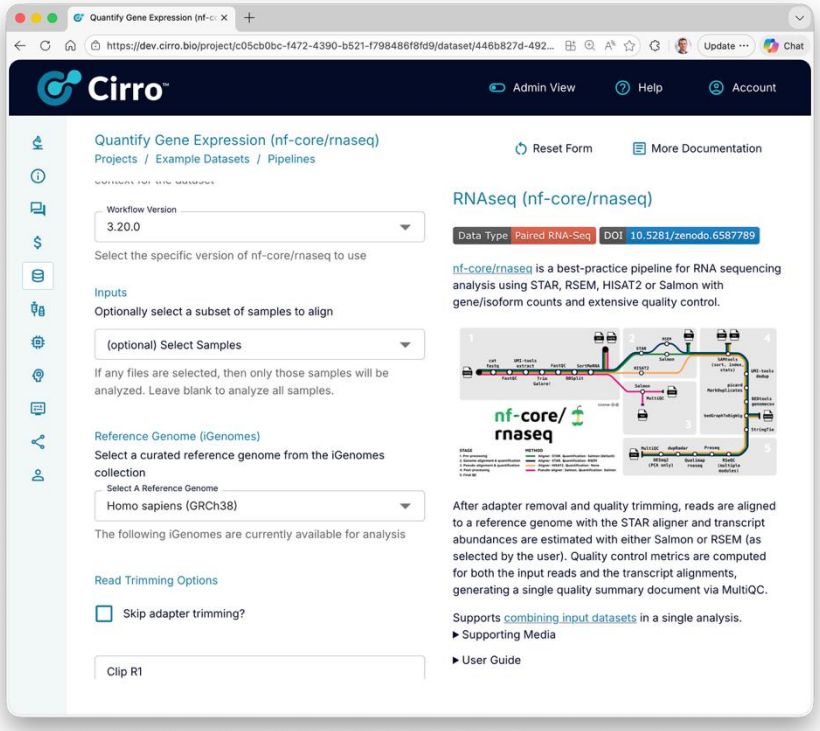
Running WDLs at Fred Hutch

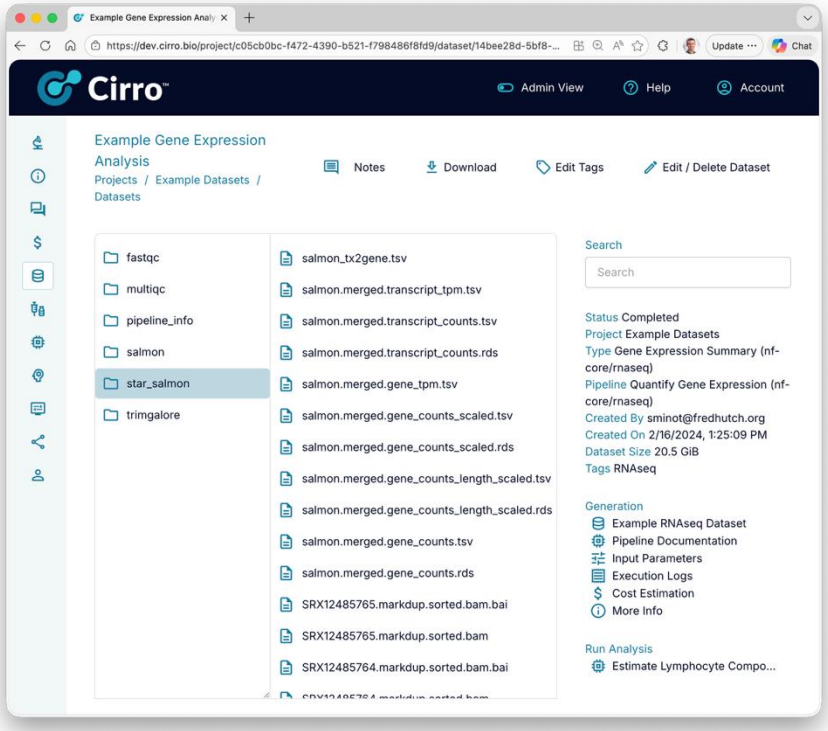
Tool	Interface	Can run	Good for
Sprocket	Command line	Any WDL	Development, running workflows manually
PROOF Workbench	Website	Any WDL	Running at scale, validating before moving to the cloud
Cirro	Website	Cirro WDLs	Using established pipelines

Also popular: miniWDL, Terra, AnVIL

Deploying Workflows in the Cirro Data Platform







1. Ingest Raw Data **2. User Friendly Pipeline Execution** **3. Pipeline Outputs + Provenance**

Tips for Success

- **Test then scale:** Start with a few samples first
- **Spot-check outputs:** Ensure they look reasonable
- **Check workflow status:** Confirm it started and didn’t fail
- **Read logs to troubleshoot:** Scan for error messages to get context

Reach out on Slack, we've probably seen your issue before

Fred Hutch WDL Resources

WILDS WDL Library
Validated WDLs

WILDS Docker Library
Validated Docker images

PROOF Workbench
Run WDLs on the cluster through a website

Fred Hutch Data Science Help

SciWiki
Useful knowledgebase

SciComp
Get help with the cluster, the cloud, data storage

FH-Data Slack
Crowdsourcing, Q&A

Data House Calls
Short meetings for personalized help

Data Science Lab Training Spring Quarter, 2026

Courses

- Bash for Bioinformatics*
- Machine Learning for Python*(new!)
- Bioconductor for Genomics*
- Intro to R

**has prerequisites*

Workshops

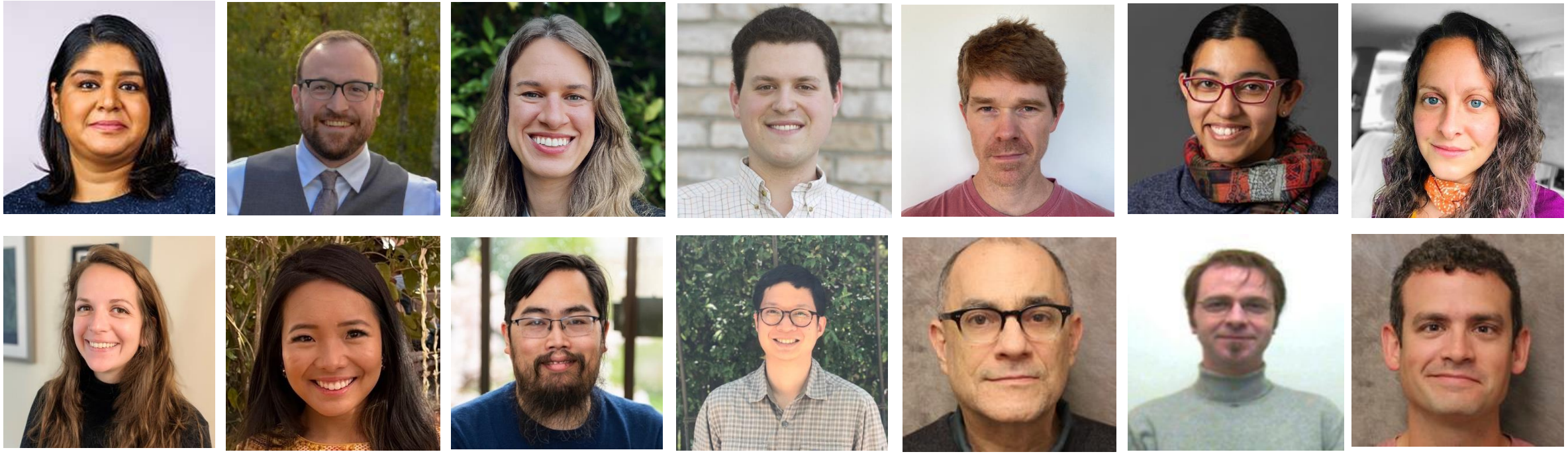
- Better Tables
- Intro to Command Line
- Intro to Cluster Computing

Registration opens soon!
Quarter begins April 28

For more info: send blank email to dasltraining-join@lists.fhcrc.org

Thank you to everyone who made this Deep Dive possible!

And thank you all for participating!!



Fred Hutch Office of the Chief Data Officer (OCDO)

Fred Hutch Scientific Computing (SciComp)

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Thank you